

```
create database alltypejoins;
use alltypejoins;
```

```
CREATE TABLE departments (
dept_id INT PRIMARY KEY,
dept_name VARCHAR(50),
location VARCHAR(50)
);
INSERT INTO departments VALUES
(10, 'HR', 'Jaipur'),
(20, 'IT', 'Bangalore'),
(30, 'Finance', 'Mumbai'),
(40, 'Marketing', 'Delhi'),
(50, 'Legal', 'Pune');
```

```
CREATE TABLE employees (
emp_id INT PRIMARY KEY,
emp_name VARCHAR(50),
dept_id INT,
salary INT,
city VARCHAR(50)
);
INSERT INTO employees VALUES
(101, 'Amit', 10, 45000, 'Jaipur'),
(102, 'Bhavna', 20, 72000, 'Bangalore'),
(103, 'Chirag', 20, 68000, 'Bangalore'),
(104, 'Divya', 30, 55000, 'Mumbai'),
(105, 'Esha', NULL, 39000, 'Delhi'),
(106, 'Farhan', 40, 61000, 'Delhi'),
(107, 'Gaurav', 60, 50000, 'Kolkata'),
(108, 'Hina', 30, 83000, 'Mumbai');
```

```
CREATE TABLE projects (
proj_id INT PRIMARY KEY,
proj_name VARCHAR(50),
dept_id INT,
budget INT
);
```

```
INSERT INTO projects VALUES
(1, 'Payroll Revamp', 10, 120000),
(2, 'Cloud Migration', 20, 500000),
(3, 'Mobile App', 20, 300000),
(4, 'Audit Automation', 30, 250000),
(5, 'Brand Refresh', 40, 180000),
(6, 'Data Lake', 70, 400000);
```

```
-- Q1. List each employee with their department name and location.
select e.emp_id , e.emp_name , d.dept_name , d.location
from employees e
inner join departments d
on e.dept_id = d.dept_id;
```

```
-- Q2. Show every project along with the name of the department that owns it.
select p.proj_id , p.proj_name , d.dept_name , p.budget
from projects p
inner join departments d
on p.dept_id = d.dept_id;
```

```

-- Q3. Show employee names together with the projects run by their own department.
select e.emp_name, d.dept_name, p.proj_name
from employees e
inner join departments d
on e.dept_id = d.dept_id
inner join projects p
on d.dept_id = p.dept_id;

-- Q4. List ALL employees with their department name; show NULL when the employee
has no matching
-- department.
select e.emp_id , e.emp_name , d.dept_id , d.dept_name
from employees e
left join departments d
on e.dept_id = d.dept_id;

-- Q5. List ALL departments with their projects; departments with no project must
still appear.
select d.dept_id , d.dept_name , p.proj_name , p.budget
from departments d
left join projects p
on d.dept_id = p.dept_id;

-- Q6. Find only those employees who do NOT belong to any valid department.
select e.emp_id , e.emp_name , e.dept_id
from employees e
left join departments d
on e.dept_id = d.dept_id
where d.dept_id is null;

-- Q7. List ALL departments and any employees in them, using a RIGHT JOIN with
employees on the left.
select e.emp_name , d.dept_id , d.dept_name
from employees e
right join departments d
on e.dept_id = d.dept_id;

-- Q8. List ALL projects and the department that owns them, keeping projects that
point to a missing department.
select d.dept_name , p.proj_id , p.proj_name , p.dept_id
from departments d
right join projects p
on d.dept_id = p.dept_id;

-- Q9. Find departments that currently have NO employee assigned.
select d.dept_id , d.dept_name , d.location
from departments d
left join employees e
on e.dept_id = d.dept_id
where e.emp_id is null;

-- Q10. Produce every possible pairing of the Finance/Legal departments with
employees earning above 70000.
select e.emp_name, e.salary, d.dept_name
from employees e
cross join departments d
where e.salary > 70000 and d.dept_name in ('Finance', 'Legal');

-- Q11. Pair every project having a budget of at least 400000 with every department

```

```
located in Mumbai or Pune.
select p.proj_name , p.budget , d.dept_name , d.location
from projects p
cross join departments d
where budget >= 400000 and d.location in ('mumbai','pune');
```

```
-- Q12. How many total row combinations result from CROSS JOIN of employees and
departments? Show the count.
select count(*) as total_combination
from employees
cross join departments ;
```

```
-- Q13. Show employees whose salary is greater than 60000.
select * from employees
where salary > 60000;
```

```
-- Q14. Show all employees based in Delhi or Mumbai.
select * from employees
where city in ('Delhi','Mumbai');
```

```
-- Q15. Show employees whose dept_id is NULL (unassigned employees).
select * from employees
where dept_id is null ;
```

```
-- Q16. Show departments located in Bangalore or Delhi.
select * from departments
where location in ('Bangalore','Delhi');
```

```
-- Q17. Show departments whose dept_id is 30 or higher.
select * from departments
where dept_id >= 30;
```

```
-- Q18. Show departments whose name starts with the letter "F" or "L".
select * from departments
where dept_name LIKE 'F%' or dept_name LIKE 'L%';
```

```
-- Q19. Show projects with a budget between 200000 and 400000 (inclusive).
select * from projects
where budget between 200000 and 400000;
```

```
-- Q20. Show all projects that belong to dept_id 20.
select * from projects
where dept_id = 20;
```

```
-- Q21. Show projects whose name contains the word "a" and budget is under 300000.
select * from projects
where proj_name like '%a%' and budget < 300000;
```

```
-- Q22. Show employee name, department name, location and project name for all
matching rows across all three tables.
select e.emp_name , d.dept_name , d.location , p.proj_name , p.budget
from employees e
join departments d
on e.dept_id = d.dept_id
join projects p
on d.dept_id = p.dept_id;
```

```
-- Q23. Show ALL employees, plus department and project details where available
(keep employees even with no dept/project).
```

```
select e.emp_id, e.emp_name, d.dept_name, p.proj_name
from employees e
left join departments d
on e.dept_id = d.dept_id
left join projects p
on d.dept_id = p.dept_id;
```

-- Q24. Show employees earning more than 60000 along with department and any project over 250000 budget.

```
select e.emp_name , e.salary , d.dept_name , p.proj_name , p.budget
from employees e
left join departments d
on e.dept_id = d.dept_id
left join projects p
on d.dept_id = p.dept_id
where salary > 60000 and budget > 250000;
```

-- Q25. List ALL departments with their employees and projects, including departments having neither.

```
select d.dept_id , d.dept_name , e.emp_name , p.proj_name
from departments d
left join employees e
on d.dept_id = e.dept_id
left join projects p
on e.dept_id = p.dept_id;
```

-- Q26. Show employees who work in a department located in Bangalore or Mumbai, along with the projects of that department.

```
select e.emp_name , d.location , p.proj_name
from employees e
left join departments d
on e.dept_id = d.dept_id
left join projects p
on d.dept_id = p.dept_id
where location IN ('bangalore','mumbai');
```

-- Q27. Pair employees who work in the same city (avoid duplicate pairs and self-pairing).

```
SELECT e1.emp_name AS employee_1 , e2.emp_name AS employee_2 , e1.city
FROM employees e1
inner join employees e2
on e1.city = e2.city and e1.emp_id < e2.emp_id;
```

-- Q28. Show every project along with employees of that department; keep projects with no employees.

```
select p.proj_name, p.dept_id, e.emp_name
from projects p
left join employees e
on p.dept_id = e.dept_id;
```